Dr. Nodali Ndraha

Yogyakarta, ID | nodali.ndraha@brin.go.id | 0852-6659-5844 | www.nodali.com | nodali | nndraha

Experience

National Research and Innovation Agency (BRIN), Junior Researcher — Yogyakarta, Indonesia — Mar 2025 – present — 1 year 4 months

Focusing on supporting the management of food safety, my research aims to develop innovative strategies and tools to ensure the quality and safety of food products. By examining various aspects of food safety management, such as risk assessment, monitoring, and regulatory compliance, I strive to enhance the effectiveness and efficiency of food safety systems. Through this work, I aim to contribute to the protection of public health and the promotion of consumer confidence in the food supply.

National Research and Innovation Agency (BRIN), Technical Policy Reviewer — Yogyakarta, Indonesia — Feb 2025 – Feb 2025 — 1 month

Responsible for evaluating scientific and technical policies, guidelines, or proposals to ensure they are aligned with national research priorities, regulatory standards, and scientific best practices. This role involves reviewing documents, analyzing technical content, providing constructive feedback, and recommending improvements to enhance policy effectiveness.

National Research and Innovation Agency (BRIN), Scientific Data Analyst — Yogyakarta, Indonesia — Mar 2024 – Jan 2025 — 1 year

Responsible for collecting, managing, and analyzing research data to support scientific studies and innovation. This role involves working with researchers to understand data needs, applying statistical and computational tools to analyze complex datasets, and creating clear reports and visualizations to communicate findings. The analyst also helps ensure data quality, supports scientific publications, and may provide training in data analysis methods. Their work supports evidence-based decision-making and contributes to the success of national research and innovation projects.

National Taiwan Ocean University, Postdoctoral researcher — Keelung, Taiwan — 2022 – 2024 — 2 years

Developed molecular methods for detecting foodborne pathogens in poultry

SNV (Stichting Nederlandse Vrijwilligers), Nutrition Sensitive Agriculture Advisor — Lombok, Indonesia — 2014 – 2015 — 1 year

Carried out a nutritional baseline study analysis and agricultural/food availability study within the selected communities.

Education

PhD **National Taiwan Ocean University**, Food Science — Keelung, Taiwan — 2017 – 2021

MS **National Taiwan Ocean University**, Food Science — Keelung, Taiwan — 2015 – 2017

BS **Universitas Sumatera Utara**, Agricultural Engineering — Medan, Indonesia — 2005 – 2009

Volunteer

NTOU International Student Association, Chairman — Keelung, Taiwan — Jan 2017 – Jan 2018

Directed the operations and initiatives of the international student association.

- Organized cultural and social events to foster community engagement among students.
- Facilitated the orientation and onboarding of new international students arriving in Taiwan.

Publications

Reducing product nonconformity through QMRA: A risk-based management approach for Listeria monocytogenes in a Taiwanese alfalfa sprout supply chain — Jan 2026
C.-H. Lin, J. Chen, N. Ndraha, H-I. Hsiao
doi.org/10.1016/j.foodcont.2026.111957

A comparison of machine learning models for predicting Vibrio parahaemolyticus in oysters — Jan 2025
Nodali Ndraha, Hsin-I Hsiao
doi.org/10.1016/j.mran.2025.100345

Assessment and Validation of Predictive Growth Models for Locally Isolated Salmonella enterica and Listeria monocytogenes in Alfalfa Sprouts — Jan 2025
N. Ndraha, C.-H. Lin, A. P. Goh, G. D. Tran, L. M. Su, C. L. Huang, C.-Q. Chen, H-I. Hsiao
doi.org/10.1111/jfs.13171

Managing the microbiological safety of tilapia from farm to consumer — Jan 2024
N. Ndraha, H.-Y. Lin, H-I. Hsiao, H.-J. Lin
doi.org/10.1111/1541-4337.70023

Rapid detection methods for foodborne pathogens based on nucleic acid amplification: Recent advances, remaining challenges, and possible opportunities — Jan 2024
N. Ndraha, H.-Y. Lin, C.-Y. Wang, H-I. Hsiao, H.-J. Lin
doi.org/10.1016/j.fochms.2023.100183

Growth-promoting and low-salt adaptation responses boosted by spermidine in Strombidium parasulcatum, a marine bacteriovorous ciliate potentially applied to live feeds for marine larviculture — Jan 2023
Hung-Yun Lin, Bo-Ying Su, Nodali Ndraha, Sheng-Fang Tsai, Kuo-Ping Chiang, Hsin-Yun Liu, Yong-Ting Kang, Wei-Yu Yeh, Che-Chia Tsao, Yi-Min Chen, Hsin-I Hsiao, Han-Jia Lin
doi.org/10.1016/j.aquaculture.2023.739616

Modeling the risk of Vibrio parahaemolyticus in oysters in Taiwan by considering seasonal variations, time periods, climate change scenarios, and post-harvest interventions — Jan 2023
N. Ndraha, H.-Y. Lin, H.-J. Lin, Hsin-I Hsiao
doi.org/10.1016/j.mran.2023.100275

The Rapid Detection of Salmonella enterica, Listeria monocytogenes, and Staphylococcus aureus via Polymerase Chain Reaction Combined with Magnetic Nanoparticles — Jan 2023
N. Ndraha, H.-Y. Lin, S.-K. Tsai, Hsin-I Hsiao, H.-J. Lin
doi.org/10.3390/foods12213895

Vibrio parahaemolyticus in seafood: recent progress in understanding influential factors at harvest and food-safety intervention approaches — Jan 2022
Nodali Ndraha, Lihan Huang, Vivian C. H. Wu, Hsin-I Hsiao
doi.org/10.1016/j.cofs.2022.100927

Predictive models for the growth of Salmonella spp., Listeria spp., and Escherichia coli in lettuce harvested on Taiwanese farms — Jan 2022
Nodali Ndraha, Ai Ping Goh, Gia Dieu Tran, Cheng-quan Chen, Hsin-I Hsiao
doi.org/10.1111/1750-3841.16236

A climate-driven model for predicting the level of Vibrio parahaemolyticus in oysters harvested from Taiwanese farms using elastic net regularized regression Jan 2022
Nodali Ndraha, Hsin-I Hsiao
doi.org/10.1016/j.mran.2022.100201

Effect of Chitosan Incorporation on the Development of Acrylamide during Maillard Reaction in Fructose–Asparagine Model Solution and the Functional Characteristics of the Resultants Jan 2022
H.-T. V. Lin, Y.-S. Ting, N. Ndraha, H-I. Hsiao, W.-C. Sung
doi.org/10.3390/polym14081565

Predictive models for the effect of environmental factors on the abundance of Vibrio parahaemolyticus in oyster farms in Taiwan using extreme gradient boosting Jan 2021
N. Ndraha, H-I. Hsiao, Y.-Z. Hsieh, A. K. Pradhan
doi.org/10.1016/j.foodcont.2021.108353

Influence of climatic factors on the temporal occurrence and distribution of total and pathogenic Vibrio parahaemolyticus in oyster culture environments in Taiwan Jan 2021
N. Ndraha, H-I. Hsiao
doi.org/10.1016/j.fm.2021.103765

Managing the risk of Vibrio parahaemolyticus infections associated with oyster consumption: A review Jan 2020
N. Ndraha, H. Wong, Hsin-I Hsiao
doi.org/10.1111/1541-4337.12557

Challenges with food waste management in the food cold chains Jan 2020
N. Ndraha, J. Vlajic, C.-C. Chang, Hsin-I Hsiao
doi.org/10.1016/B978-0-12-817121-9.00022-X

The risk assessment of Vibrio parahaemolyticus in raw oysters in Taiwan under the seasonal variations, time horizons, and climate scenarios Jan 2019
N. Ndraha, H-I. Hsiao
doi.org/10.1016/j.foodcont.2019.03.020

Exposure assessment and sensitivity analysis for chilled shrimp during distribution: A case study of home delivery services in Taiwan Jan 2019
N. Ndraha, H-I. Hsiao
doi.org/10.1111/1750-3841.14498

Evaluation of the cold chain management options to preserve the shelf life of frozen shrimps: A case study in the home delivery services in Taiwan Jan 2019
N. Ndraha, W.-C. Sung, H-I. Hsiao
doi.org/10.1016/j.jfoodeng.2018.08.010

Time-temperature abuse in the food cold chain: Review of issues, challenges, and recommendations Jan 2018
N. Ndraha, H-I. Hsiao, J. Vlajic, M.-F. Yang, H.-T. V. Lin
doi.org/10.1016/j.foodcont.2018.01.027

Comparative study of imported food control systems of Taiwan, Japan, the United States, and the European Union Jan 2017
Nodali Ndraha, Hsin-I Hsiao, Wayne Chung Chih Wang
doi.org/10.1016/j.foodcont.2017.02.051

Skills

QMRA, R for food science

Languages

Indonesian
Native speaker

English
Professional

Interests

Food Microbiology

Certificates

Preventive Controls Qualified Individual Aug 2018

Awareness Training for Hazard Analysis and Critical Control Point (HACCP) May 2025

Awareness Training for Good Manufacturing Practice (GMP) May 2025

Awareness Training for Food Safety Management Based on ISO 22000:2018 May 2025

Awareness Training for Sanitation Standard Operating Procedure (SSOP) May 2025

Projects

Microbiological Food Safety of Green Vegetables Jan 2025 – Jan 2027

This project proposes to develop predictive models and quantitative microbial risk assessments for pathogens on green vegetables produced by local farmers in Central Java, Indonesia, specifically focusing on the impacts of climate change to enhance food safety and public health.

- Development of molecular methods for detecting foodborne pathogens in green vegetables
- Development of predictive modeling for the occurrence and level of foodborne pathogens in green vegetables

STAPHYLO-GUARD Jan 2025 – Jan 2027

This project develops a rapid test kit for detecting Staphylococcus aureus in free meal programs.

- Development of molecular methods for detecting foodborne pathogens in free meal program

References

Professor Hsin-I Hsiao

Dr. Satriyo K. W.